

Software Requirements Specification

For

Refugee Relief Camp Management System

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Anshrah Shafique	20/04/2026	Document Created	1.0
Erfa Kashif	24/04/2026	System Features added	1.1
Daneya Shafiq	30/04/2026	System Features added	1.1
Anshrah Shafique	03/05/2026	Use case Model added	1.2
Erfa Kashif	03/05/2026	ERD added	1.3
Daneya Shafiq	04/05/2026	Activity Diagram added	1.4

1. Introduction

1.1 Purpose

This document provides a detailed description of the requirements for the Refugee Relief Camp Management System, including system functionalities, constraints, and external interfaces.

1.2 Document Conventions

1. Font: Arial
2. Size: 10 (body), 12 (sub-headings), 14 (headings)
3. 1.3 Abbreviations:
4. SRS: Software Requirement Specification
5. ERD: Entity Relationship Diagram
6. UI: User Interface
7. DB: Database

1.3 Intended Audience and Reading Suggestions

1. Developers
2. Project supervisors
3. Camp administrators
4. NGOs and organizations

1.4 Product Scope

The system will:

1. Register refugees, volunteers, and doctors
2. Manage shelters and camps
3. Track inventory and donations
4. Maintain medical records
5. Record incidents

Not In Scope:

1. AI-based predictions
2. GPS tracking
3. Government ID integration

1.5 References

1. Software Engineering Course Material
2. Database Design Concepts

2. Overall Description

2.1 Product Perspective

Relief camps often rely on manual systems, causing inefficiencies in managing refugees, medical records, and resources. This system aims to digitize operations and improve coordination.

2.2 Product Functions

Main modules:

1. Refugee Management
2. Shelter Management
3. Medical Management
4. Inventory Management
5. Incident Reporting
6. Donation Management

Use Case: UC-01 —Refugee Registration

Use Case Id	UC-01
Actors	Admin, Coordinator
Feature	Refugee Registration
Pre-condition	User must be logged in

Step #	Action	Software Reaction
1	Enter refugee data	Validate input
2	Assign Shelter	Update Capacity
3	View Incident Report	System fetches record
4	Enter diagnosis & treatment	System saves medical report
5	Enter Incident Details	System validates the incident report
6	Submit incident report	System saves the report and notifies the doctor
7	Views Inventory Stock	System displays current stock
8	Update inventory Quantities	System saves changes
9	Enter donation details	System validates donation information
10	Submit the donation	System stores the record and updates inventory

Post Conditions	Description
Steps 1–2	Refugee successfully registered and assigned to the shelter
Steps 3–4	Medical Record is saved and linked with the refugee
Steps 5–6	Incident Report is recorded and notifications are saved
Steps 7–8	Inventory is updated and reflects the latest stock

Steps 9–10	Donation record is stored and inventory is updated
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Use Case: UC-02 — Donor making Donation

Use Case Id	UC-02
Actors	Donor
Feature	Donation Management
Pre-condition	-

Step #	Action	Software Reaction
1	Enter donation details	System validates donation information
2	Submit the donation	System stores the record and updates inventory

Post Conditions	Description
Steps 2	Donation successfully recorded in database

Use Case: UC-03 — Shelter Assignment

Use Case Id	UC-03
Actors	Coordinator, Admin
Feature	Shelter Management
Pre-condition	User logged in .Refugee Must be registered

Step #	Action	Software Reaction
1	View available shelters	System fetches shelter data

2	Select and assign shelter to refugee	System stores the record and updates refugee data and capacity of shelter
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Post Conditions	Description
Steps 1-2	Shelter and refugee data successfully recorded in database

Use Case: UC-04 — Volunteer incident report

Use Case Id	UC-04
Actors	Volunteer
Feature	Incident Reporting
Pre-condition	Volunteer logged in

Step #	Action	Software Reaction
1	Enter incident details	System validates donation information
2	Submit the incident	System stores the record

Post Conditions	Description
Steps1- 2	Incident successfully recorded in database

Use Case: UC-05 — Inventory Management

Use Case Id	UC-05
Actors	Coordinator
Feature	Inventory Management
Pre-condition	Coordinator logged in

Step #	Action	Software Reaction
1	View Stock	Display data
2	Update Inventory	Save changes

Post Conditions	Description
Steps 1-2	Inventory successfully recorded in database

2.3 User Classes and Characteristics

The following user classes are identified for the Refugee Relief Camp Management System:

User Class	Characteristics
Camp Coordinators	Primary administrators who manage camp operations, assign camps, and oversee all modules. High frequency of use, full system access.
Admin	Government or official administrators overseeing coordinators, doctors, donations, aid distribution, refugees, and shelter/camp operations
Volunteers	Field workers who assist with refugee registration and shelter assignment. Moderate frequency of use with limited access.
Doctors	Medical personnel managing medical records and incident reports. Regular use of medical and incident modules.
NGOs / Organizations	External stakeholders who manage donations and view camp reports. Periodic use with restricted access.
System Developers	Technical staff responsible for system maintenance and updates. Administrative access for configuration and support.

2.4 Operating Environment

1. OS: Windows
2. Platform: Web Application
3. Database: Oracle
4. Browser: Chrome

2.5 Design and Implementation Constraints

1. Limited infrastructure in camps
2. User skill limitations
3. Academic deadline
4. Data privacy requirements

2.6 User Documentation

1. User manual
2. Help guide

2.7 Assumptions and Dependencies

1. Users have basic computer knowledge
2. Internet access available
3. Database server functional

3. External Interface Requirements

3.1 User Interfaces

The system will provide a web-based graphical user interface accessible via standard browsers (Chrome). Interfaces will include role-specific dashboards for coordinators, volunteers, doctors, and NGOs, with standardized navigation menus, form-based data entry screens, and clear error message displays.

3.2 Hardware Interfaces

1. Computers / Laptops
2. Servers for database

3.3 Software Interfaces

1. Database system (MySQL/Oracle)
2. Operating system (Windows/Linux)

3.4 Communications Interfaces

The system requires the following communications interfaces:

1. Internet connectivity for web-based access
2. HTTP/HTTPS protocols for secure client-server communication
3. Standard web browser communication support (Chrome)
4. Server-side API endpoints for data exchange between frontend and Oracle database

4. System Features

4.1 Refugee Registration

4.1.1 Description and Priority

Allows coordinators to register refugees, assign shelters, and maintain refugee profiles. Priority: High.

4.1.2 Stimulus/Response Sequences

1. User enters refugee data → System validates input
2. User assigns shelter → System updates capacity

4.1.3 Functional Requirements

REQ-1: System shall allow coordinators and admin to enter and save refugee personal data.

REQ-2: System shall assign refugees to available shelters and camps, and update capacity accordingly.

REQ-3: System shall display an error if invalid or incomplete refugee data is submitted.

REQ-4: System shall display refugee profile to coordinator and admin.

4.2 Medical Record Management

4.2.1 Description and Priority

Enables doctors to view refugee records, enter diagnoses and treatment details, and flag critical patients. Priority: High.

4.2.2 Stimulus/Response Sequences

1. Doctor, coordinator, and admin views refugee record → System fetches record
2. Doctor enters diagnosis and treatment → System saves medical report
3. Patient marked as critical → System saves record and notifies doctor

4.2.3 Functional Requirements

REQ-5: System shall allow doctors, coordinators, and admin to retrieve refugee medical records.

REQ-6: System shall save diagnosis and treatment details linked to the refugee.

REQ-7: System shall notify the doctor when a patient is marked critical.

4.3 Incident Reporting

4.3.1 Description and Priority

Allows authorized users to enter and submit incident reports. Priority: Medium.

4.3.2 Stimulus/Response Sequences

1. System User enters incident details → System validates the incident report
2. System User submits incident report → System saves the report and notifies the coordinator and admin

4.3.3 Functional Requirements

REQ-8: System shall validate incident report data before saving.

REQ-9: System shall notify relevant personnel upon incident report submission.

REQ-10: System shall reject invalid or incomplete incident data and prompt for correction.

4.4 Inventory Management

4.4.1 Description and Priority

Enables volunteers and coordinators to view and update inventory stock levels. Priority: Medium.

4.4.2 Stimulus/Response Sequences

1. User views inventory stock → System displays current stock
2. User updates inventory quantities → System saves changes

4.4.3 Functional Requirements

REQ-11: System shall display current inventory levels to authorized users.

REQ-12: System shall save valid inventory updates.

REQ-13: System shall reject invalid inventory updates.

REQ-14: System shall deduct distributed aid from inventory.

4.5 Donation Management

4.5.1 Description and Priority

Allows NGOs and donors to enter and submit donation details. Priority: Medium.

4.5.2 Stimulus/Response Sequences

1. User enters donation details → System validates donation information
2. User submits donation → System stores the record and updates inventory

4.5.3 Functional Requirements

REQ-13: System shall validate donation details before saving.

REQ-14: System shall update inventory records upon successful donation submission.

REQ-15: System shall show a validation error and prevent submission for invalid donation details.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

1. Fast data retrieval
2. Support multiple concurrent users
3. Efficient database queries

5.2 Safety Requirements

1. Prevent data loss
2. Backup mechanisms
3. Error handling

5.3 Security Requirements

1. User authentication
2. Role-based access control
3. Data privacy protection

5.4 Software Quality Attributes

The system shall prioritize the following quality attributes:

1. Usability: Intuitive interfaces suitable for users with basic computer knowledge
2. Reliability: Consistent availability of core features during camp operations
3. Maintainability: Modular design to allow updates and enhancements
4. Portability: Web-based platform accessible from standard browsers

5.5 Business Rules

1. Only authenticated users may access the system
2. Each user role (coordinator, doctor, volunteer, NGO) has defined access privileges
3. Refugees must be registered before any medical records or incidents can be linked to them
4. Inventory updates require authorization from coordinators or volunteers

6. Other Requirements

The following additional requirements apply to the Refugee Relief Camp Management System:

1. Database Requirements: The system shall use Oracle as the primary database. All data shall be stored with referential integrity enforced through foreign key constraints.
2. Internationalization: The system interface shall be in English. Future versions may support additional languages based on deployment region.
3. Legal Requirements: The system shall comply with applicable data privacy regulations governing the storage and processing of personal information for refugees.
4. Reuse Objectives: Common components such as authentication and role-based access control shall be designed for reuse across potential future systems.

Appendix A: Glossary

Term / Acronym	Definition
SRS	Software Requirements Specification
ERD	Entity Relationship Diagram
UI	User Interface
DB	Database
UC	Use Case
NGO	Non-Governmental Organization
HTTP/HTTPS	HyperText Transfer Protocol / Secure

Appendix B: Analysis Models

The Object-Oriented Analysis (OOA) model and the Use Case Model (UC-01) included in Section 4 serve as the primary analysis models for this document. The system is analyzed through interacting objects and user functionalities to provide a clear representation of system behavior and requirements. Additional Entity-Relationship Diagrams (ERDs), Data Flow Diagrams (DFDs), sequence diagrams, and other supporting models will be documented separately as part of the system design phase.

Fig 1: Entity Relationship Diagram

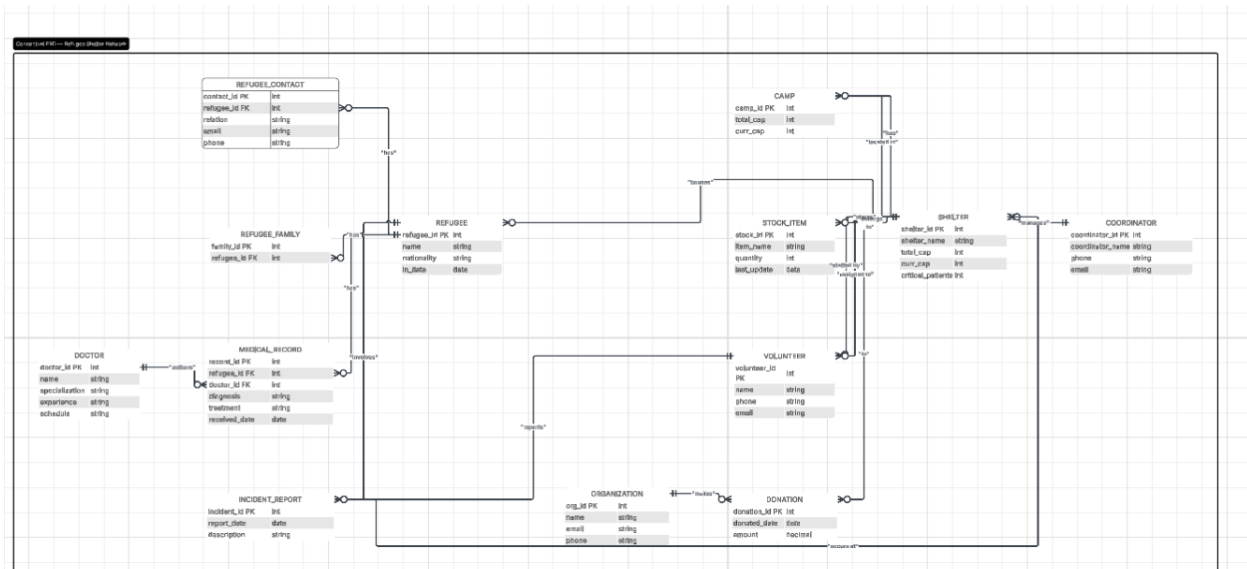


Fig 2: Class Diagram

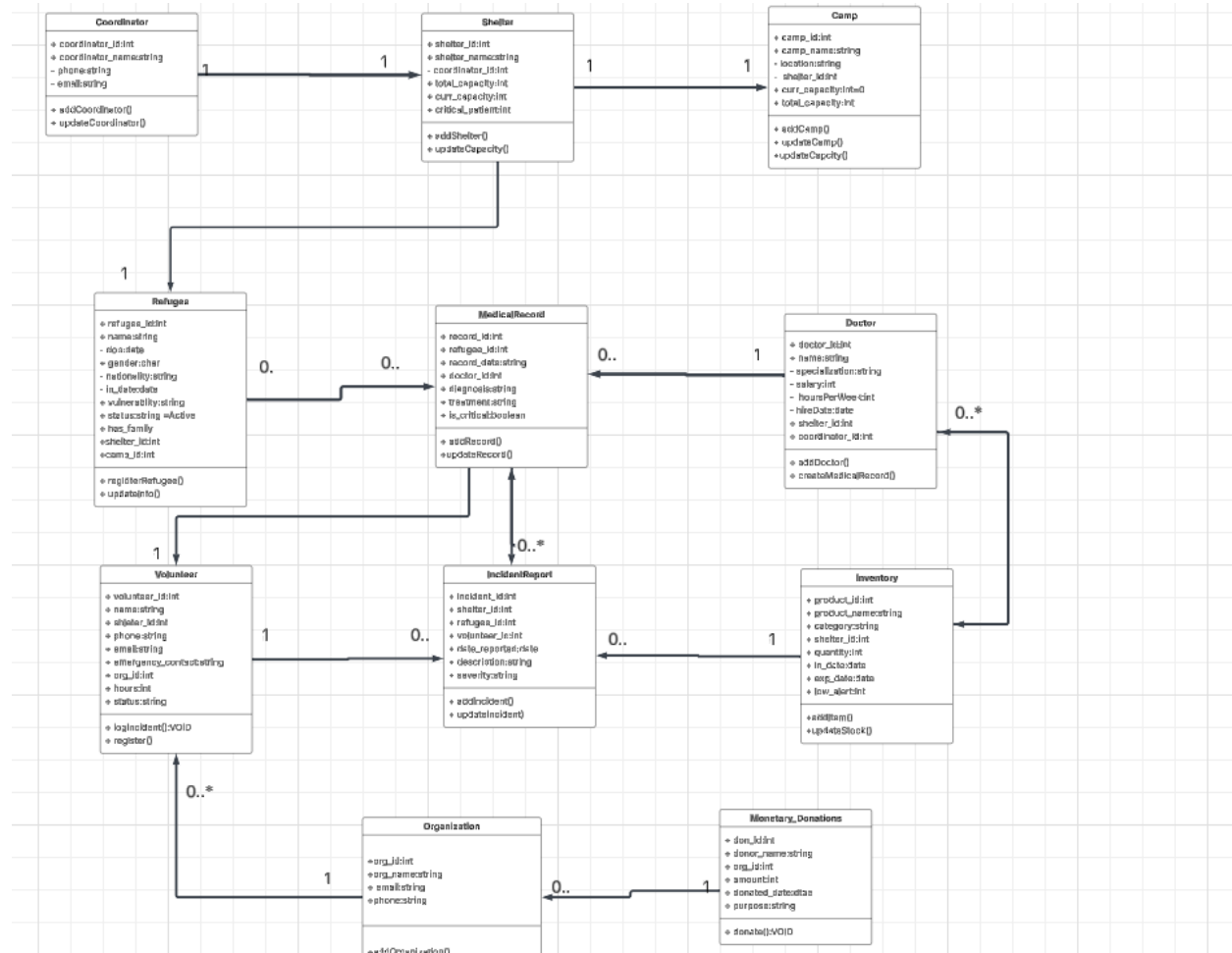
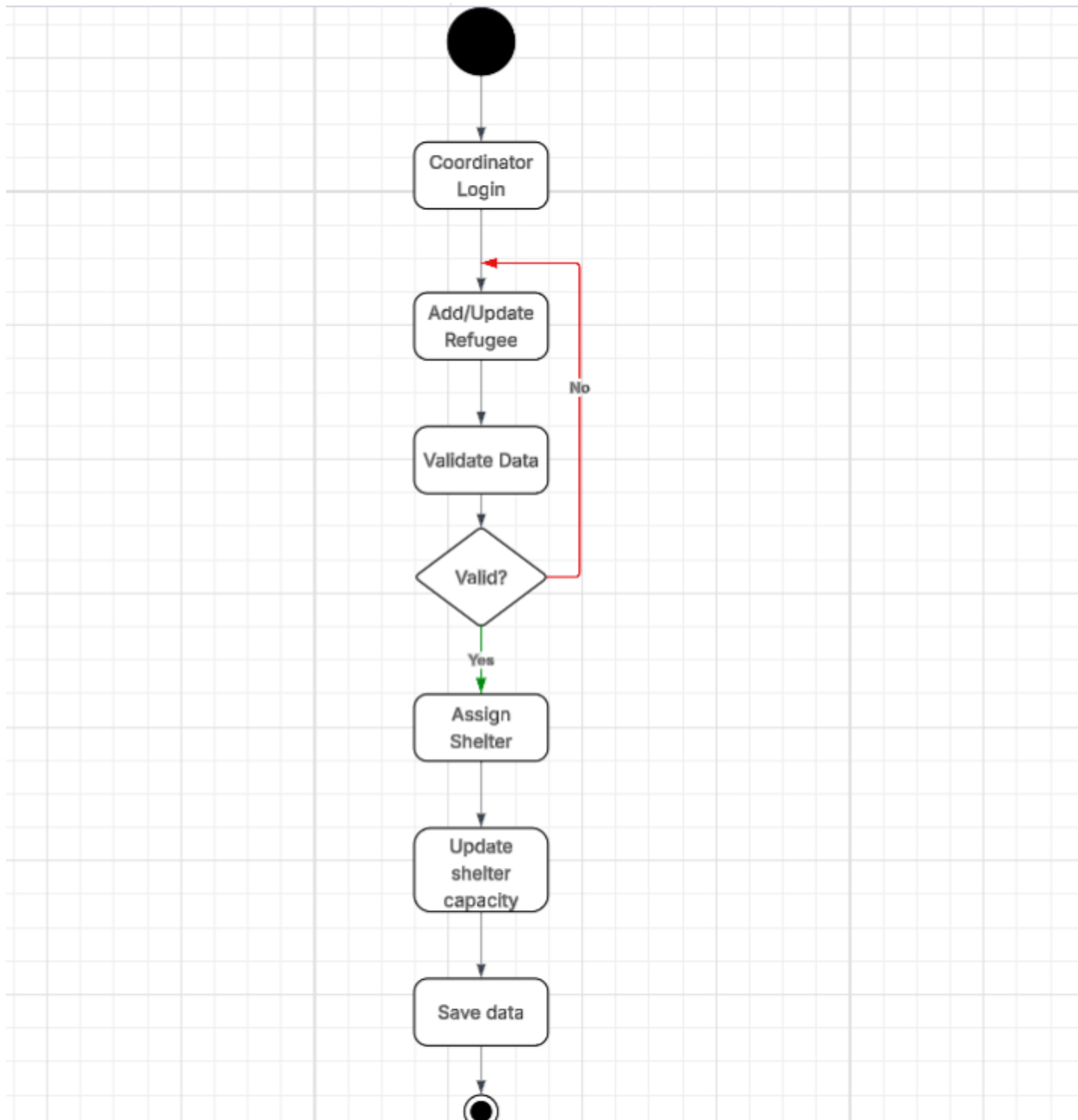
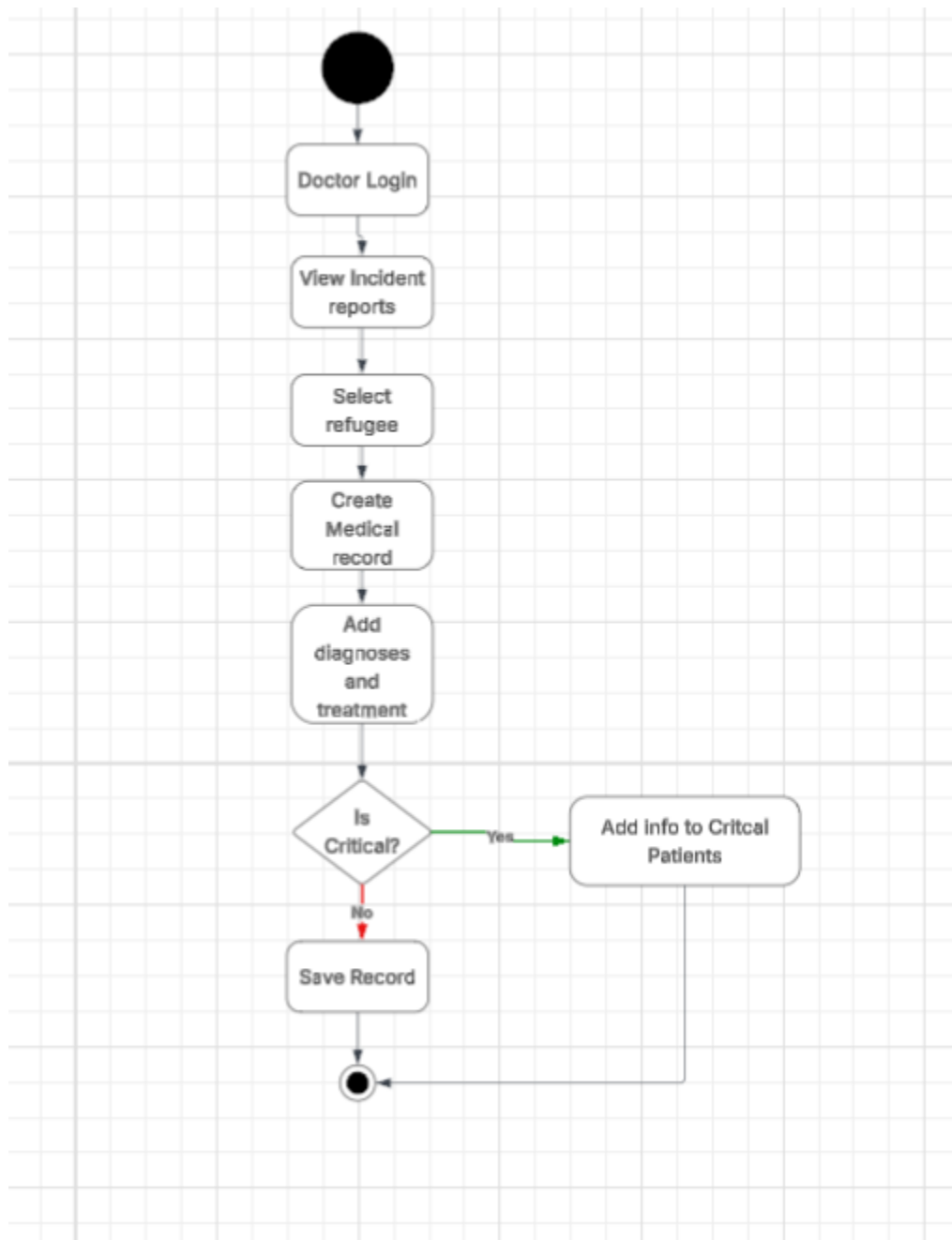


Fig 3: State Diagram

Fig 3.1: <Coordinator + System>



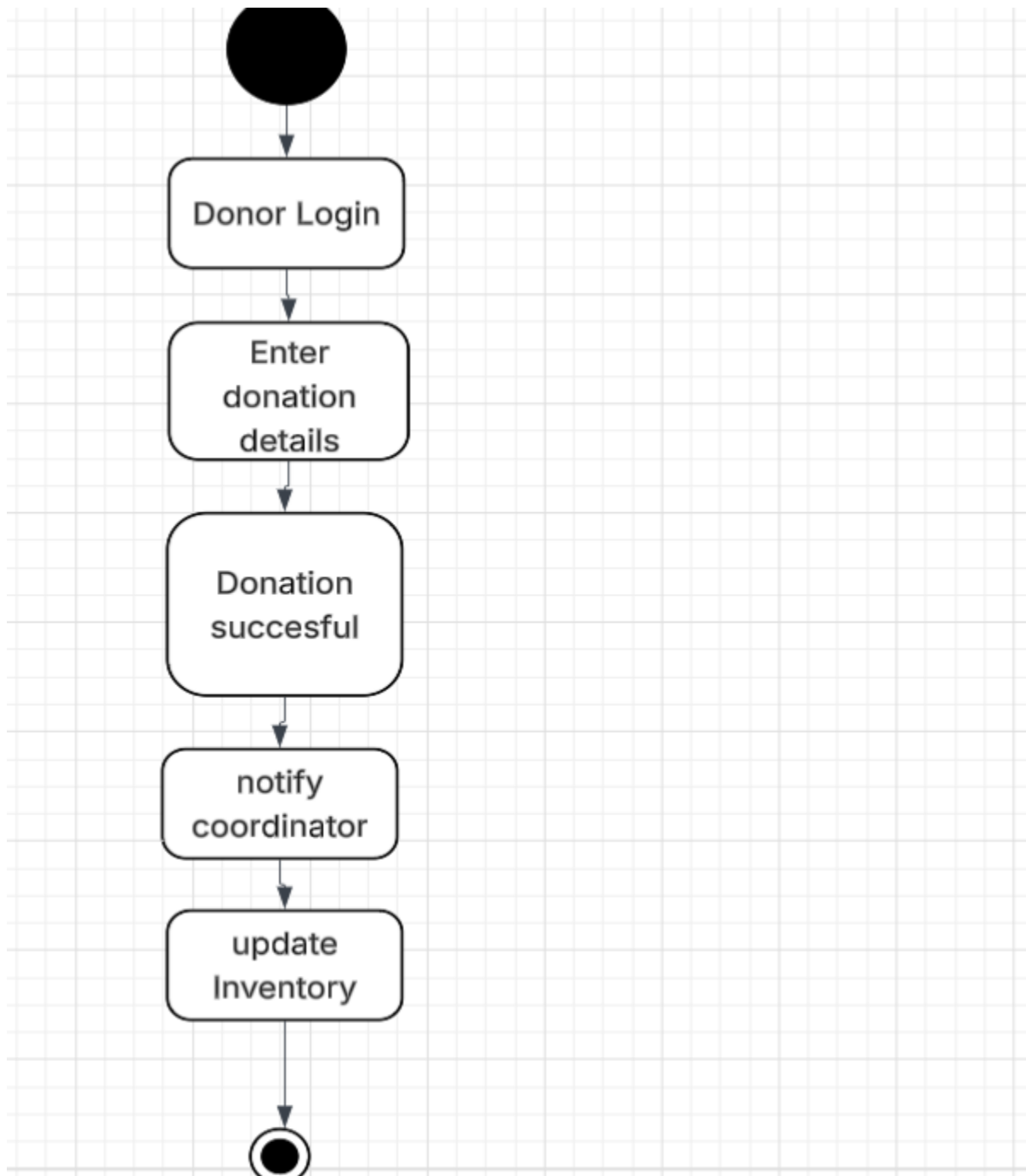
The process begins when the coordinator logs into the system. After login, they either add a new refugee or update an existing one by entering personal details like name, nationality, gender, and vulnerability status. The system then validates all the entered data to make sure nothing is missing or incorrect. Once validated, the system assigns an appropriate shelter to the refugee based on availability. The shelter's current capacity count is then automatically updated to reflect the new addition. Finally, all data is saved to the database and the process ends.

Fig 3.2: <Doctor>

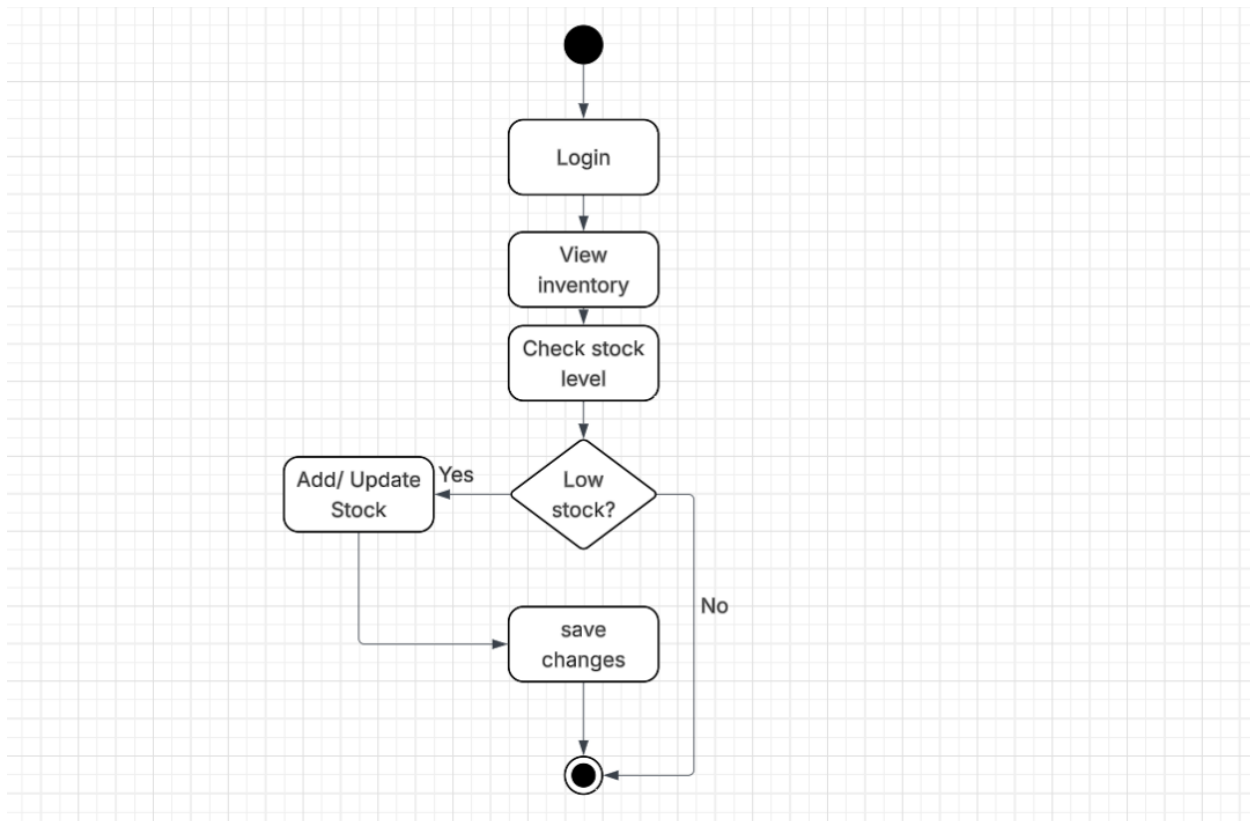
The doctor logs into the system. Instead of directly searching for a refugee, the doctor first views the list of incident reports - meaning medical records are created in response to reported incidents. The doctor then selects the specific refugee from that incident list. A new medical record is created for that refugee, and the doctor fills in the diagnosis and treatment details. The completed record is then saved to the database.

Fig 3.3: <Volunteer>

The volunteer logs in and selects the refugee or shelter that the incident involves. They then enter the full details of the incident - description, severity, date. The report is submitted and the system validates it to check all required fields are filled. Once validated, the incident is saved to the database.

Fig 3.4: <Donor>

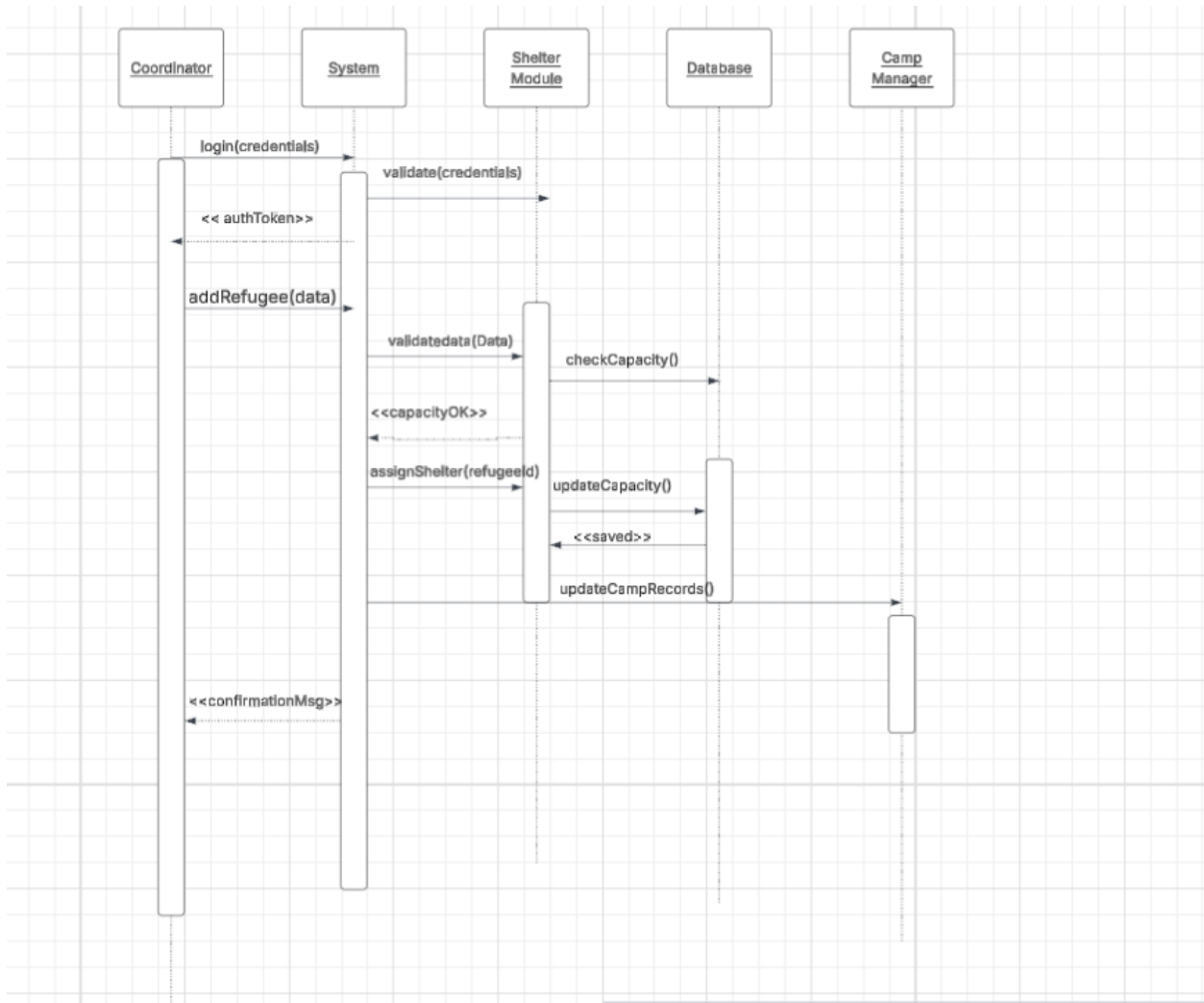
The donor logs into the system and enters the donation details such as amount, purpose, and organization information if any. The system processes and confirms the donation as successful, updating inventory, notifying the coordinator, and sending a receipt and the process ends.

Fig 3.5: <Coordinator + inventory manager>

The coordinator logs in and views the current inventory list. The system checks the stock level of all items. A decision is made — if stock is low, the coordinator adds or updates the stock quantities and the changes are saved. If stock is not low, no changes are needed and the process ends immediately.

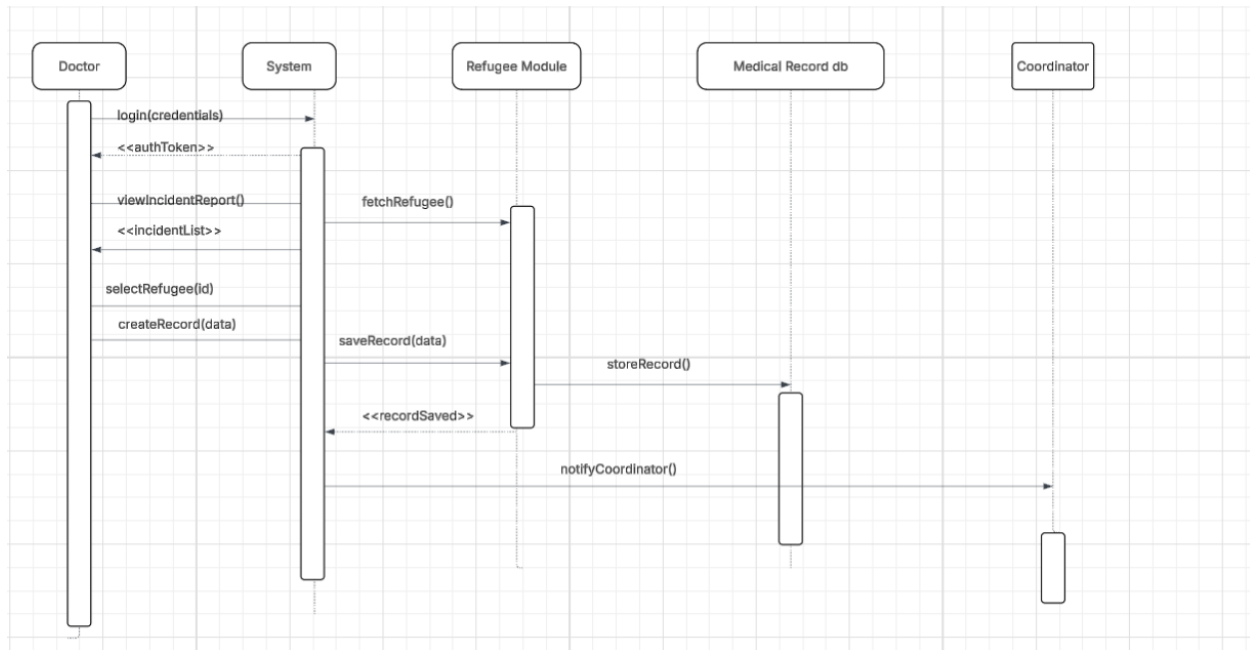
Fig 4: Sequence Diagram

Fig 4.1: <Coordinator registers a refugee >

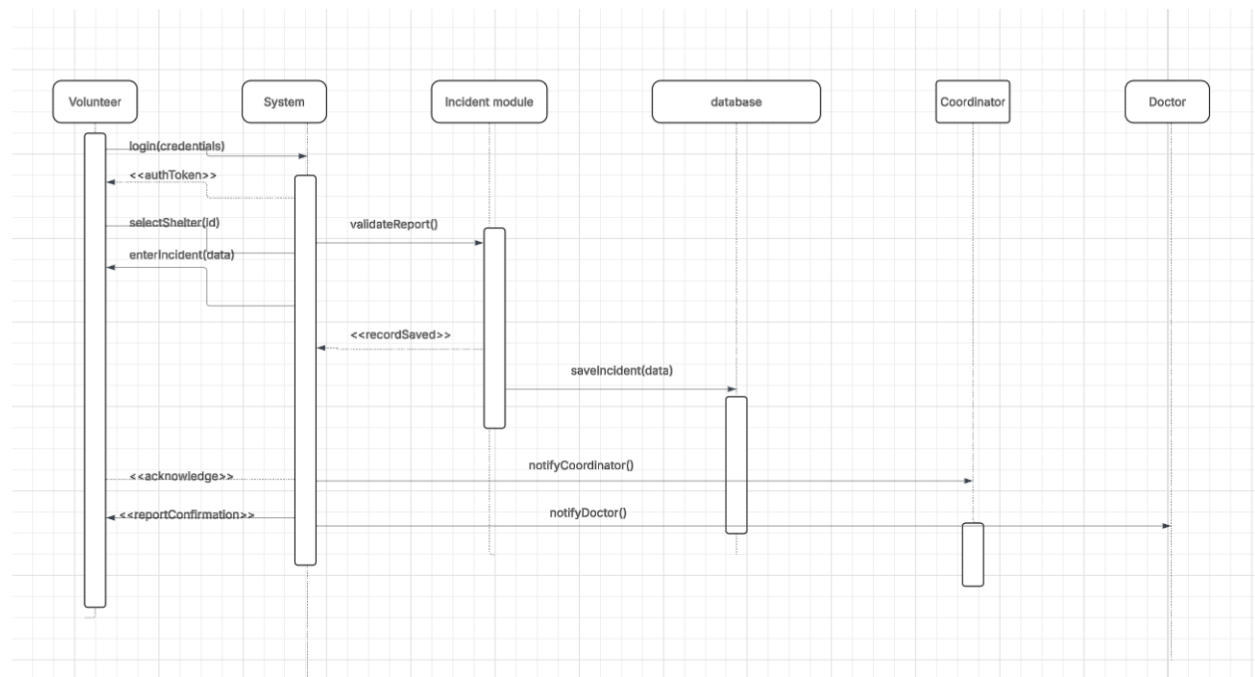


Flow:

1. System starts (filled black circle)
2. Coordinator logs in
3. Coordinator adds or updates refugee information
4. System validates the data entered
5. System assigns a shelter to the refugee
6. System updates the shelter's current capacity count
7. System saves the data to the database
8. Process ends (circle with dot)

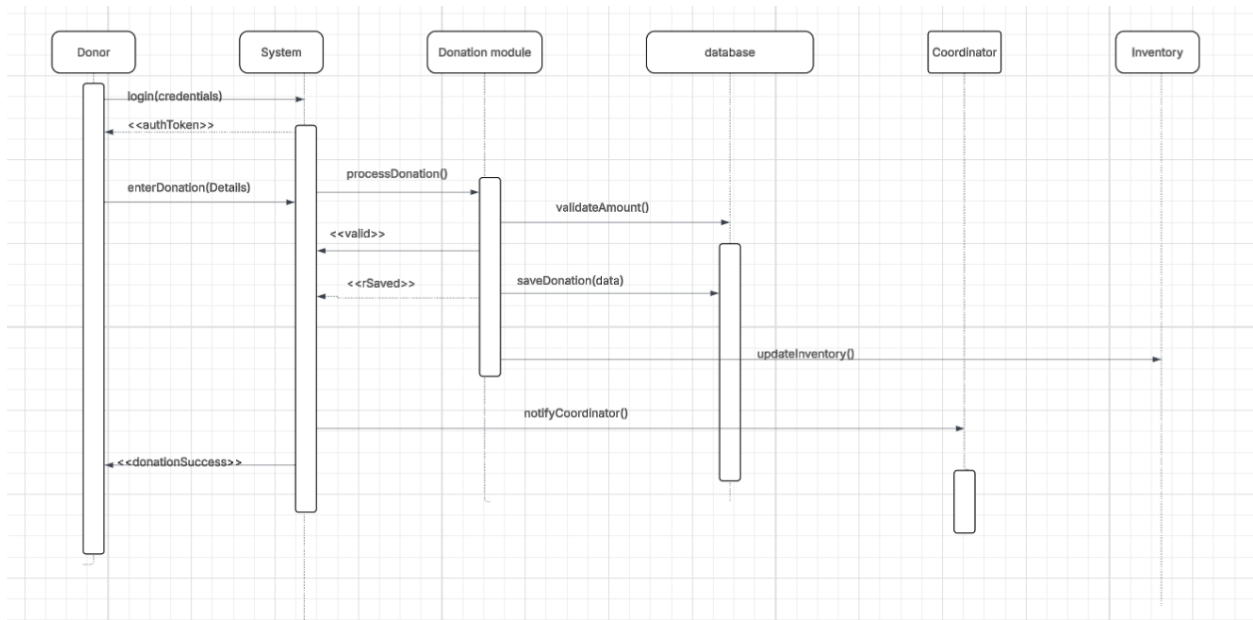
Fig 4.2: <Doctor creates medical record >**Flow:**

1. System starts
2. Doctor logs in
3. Doctor views the list of incident reports
4. Doctor selects a specific refugee from the list
5. Doctor creates a new medical record for that refugee
6. Doctor adds diagnosis and treatment details
7. System saves the record to the database
8. Process ends

Fig 4.3: <Volunteer reports an incident>**Flow:**

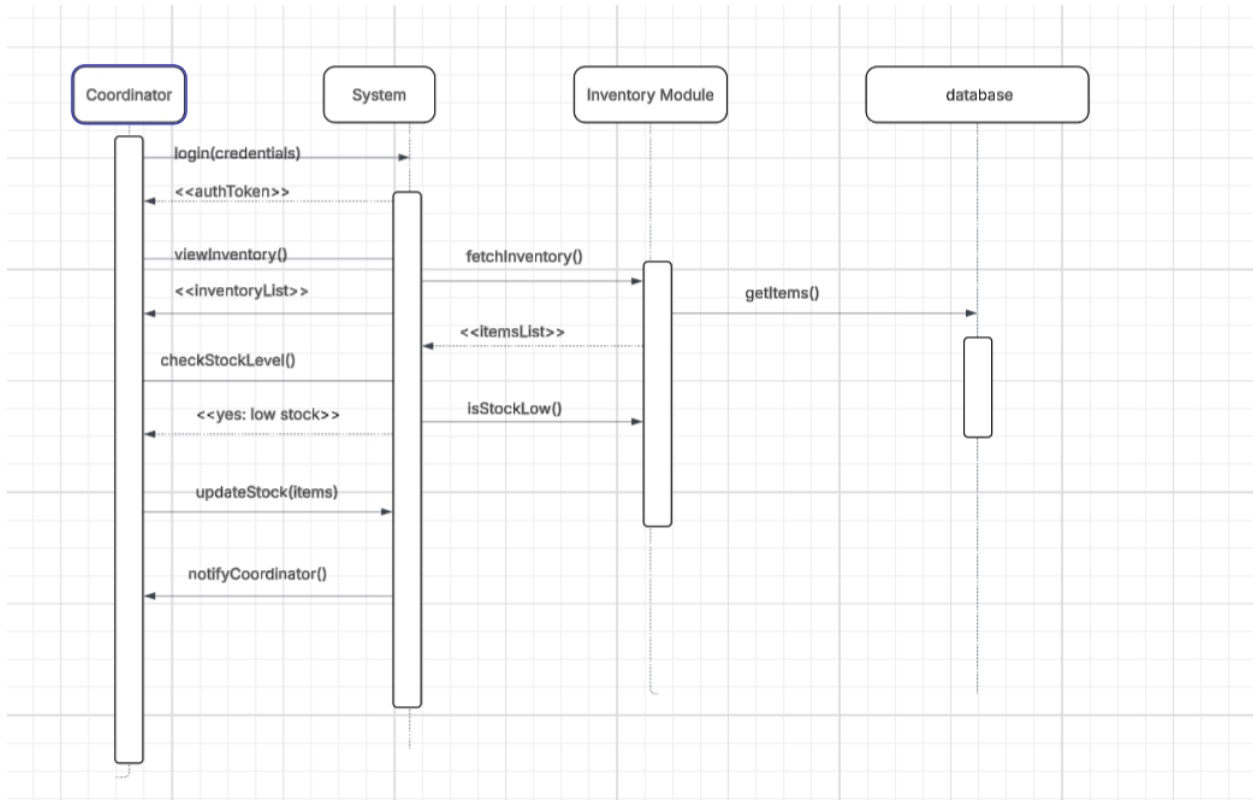
1. System starts
2. Volunteer logs in
3. Volunteer selects the refugee and/or shelter involved
4. Volunteer enters the incident details
5. Volunteer submits the report
6. System validates the report
7. System saves the incident record
8. Process ends

Fig 4.4: <Donor makes donation>



Flow:

1. System starts
2. Donor logs in
3. Donor enters donation details (amount, purpose, etc.)
4. System confirms donation successful
5. Process ends

Fig 4.5: <Coordinator manages inventory>**Flow:**

1. System starts
2. User logs in
3. User views the inventory list
4. System checks current stock levels
5. **Decision: Is stock low?**
 - a. **Yes** → User adds or updates stock quantities → System saves changes → End
 - b. **No** → System goes directly to end (no changes needed)

Fig 5: Traceability Matrix

Requirements Traceability Matrix						
Refugee Relief Camp Management System SRS v1.0						
Req. ID	Requirement Description	UC-01	UC-02	UC-03	UC-04	UC-05
Refugee Registration (4.1)						
REQ-1	Register/save refugee personal data	✓				
REQ-2	Assign refugee to shelter, update capacity	✓	✓	✓		
REQ-3	Error on invalid refugee data	✓				
REQ-4	Display refugee profile	✓				
Medical Record Mgmt (4.2)						
REQ-5	Retrieve refugee medical records	✓				
REQ-6	Save diagnosis & treatment details	✓				
REQ-7	Notify doctor on critical patient	✓				
Incident Reporting (4.3)						
REQ-8	Validate incident report data	✓			✓	
REQ-9	Notify personnel on incident submission	✓			✓	
REQ-10	Reject invalid incident data	✓			✓	
Inventory Mgmt (4.4)						
REQ-11	Display current inventory levels	✓				✓
REQ-12	Save valid inventory updates	✓				✓
REQ-13	Reject invalid inventory updates / Validate donation	✓				✓
REQ-14	Deduct distributed aid / Update inventory on donation	✓	✓			✓
Donation Mgmt (4.5)						
REQ-15	Show validation error for invalid donation					✓

✓ = Requirement traced to Use Case

Fig 6: Interaction Matrix

Interaction Matrix						
Refugee Relief Camp Management System SRS v1.0						
Part A — Actor × System Module Interactions						
Actor / Module	Refugee Mgmt	Shelter Mgmt	Medical Record	Incident Report	Inventory Mgmt	Donation Mgmt
Admin	R/W	R/W	R/W	R/W	R/W	R/W
Coordinator	R/W	R/W	R	R/W	R/W	R
Doctor	R		R/W	R		
Volunteer	R	R		W	R/W	
Donor / NGO					R	W

Part B — Module × Module Interactions						
Module	Refugee Mgmt	Shelter Mgmt	Medical Record	Incident Report	Inventory Mgmt	Donation Mgmt
Refugee Mgmt	—	●	●	●		
Shelter Mgmt	●	—		●		
Medical Record	●		—	●		
Incident Report	●	●	●	—		
Inventory Mgmt					—	●
Donation Mgmt					●	—

● R/W = Read & Write
● R = Read only
● W = Write only
● = Modules

Appendix C: To Be Determined List

The following items remain to be determined and will be tracked to closure:

- TBD-01: Specific backup schedule and recovery time objectives for safety requirements.
- TBD-02: Detailed UI wireframes and screen layout specifications.
- TBD-03: Full use case details for UC-02 through UC-05 (Medical Record Management, Incident Reporting, Inventory Management, and Donation Management).